5001 Homework 4

LAN Tianwei
tlanaa@connect.ust.hk

December 17, 2025

Q1: Practicing with Quantum System

(a)

(i) The state $|\psi\rangle = N(2|0\rangle - |1\rangle)$ must satisfy $\langle\psi|\psi\rangle = 1$. Compute the norm:

$$\|2|0\rangle - |1\rangle\|^2 = |2|^2 + |-1|^2 = 4 + 1 = 5.$$

Thus, the normalization constant is

$$N = \frac{1}{\sqrt{5}}.$$

(ii) In quantum mechanics, an observable must be represented by a Hermitian operator. The given operator is

$$A = \lambda|0\rangle\langle 0| + \epsilon|0\rangle\langle 1| + \epsilon|1\rangle\langle 0| + \lambda|1\rangle\langle 1|,$$

which has the matrix representation

$$A = \begin{bmatrix} \lambda & \epsilon \\ \epsilon & \lambda \end{bmatrix}.$$

For A to be Hermitian, we require $A^\dagger = A$ Since the off-diagonal elements are equal, this holds if and only if both λ and ϵ are real. Thus, the necessary condition is:

$$\lambda \in \mathbb{R}, \quad \epsilon \in \mathbb{R}.$$

The eigenvalues of $A = \begin{bmatrix} \lambda & \epsilon \\ \epsilon & \lambda \end{bmatrix}$ are

$$\mu_1 = \lambda + \epsilon, \quad \mu_2 = \lambda - \epsilon,$$

with corresponding *normalized* eigenvectors

$$|\phi_1\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad |\phi_2\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

(iii) Given ψ in (i), the probabilities of measuring each eigenvalue are:

$$P(\mu_1) = |\langle\phi_1|\,\psi\rangle|^2 = \left|\frac{1}{\sqrt{2}} \cdot \frac{2}{\sqrt{5}} + \frac{1}{\sqrt{2}} \cdot \frac{-1}{\sqrt{5}}\right|^2 = \left|\frac{1}{\sqrt{10}}\right|^2 = \frac{1}{10},$$

$$P(\mu_2) = |\langle\phi_2|\,\psi\rangle|^2 = \left|\frac{1}{\sqrt{2}} \cdot \frac{2}{\sqrt{5}} + \frac{-1}{\sqrt{2}} \cdot \frac{-1}{\sqrt{5}}\right|^2 = \left|\frac{3}{\sqrt{10}}\right|^2 = \frac{9}{10}.$$

Thus, the probabilities are $\frac{1}{10}$ for $\lambda + \epsilon$ and $\frac{9}{10}$ for $\lambda - \epsilon$.

(b)

Consider the two-qubit state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle).$$

In computational basis, this corresponds to coefficients:

$$a = 0, \quad b = \frac{1}{\sqrt{2}}, \quad c = -\frac{1}{\sqrt{2}}, \quad d = 0.$$

A two-qubit state is separable if and only if $ad = bc$. Compute:

$$ad = 0 \cdot 0 = 0, \quad bc = \left(\frac{1}{\sqrt{2}}\right)\left(-\frac{1}{\sqrt{2}}\right) = -\frac{1}{2}.$$

Since $ad \neq bc$, the state is **entangled** and cannot be written as a tensor product of two single-qubit states.

(c)

If the state $|\psi\rangle = \sum_{i,j} \psi_{ij} |e_i\rangle |e_j\rangle$ is unentangled, then it can be written as a product state:

$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle,$$

which implies that the coefficient matrix $\Psi = [\psi_{ij}]$ has rank 1.

For any $n \times n$ matrix with rank less than n (here, rank $= 1$), its determinant is 0.

Thus, if $|\psi\rangle$ is unentangled, then:

$$\det(\Psi) = 0.$$

(d)

From (c), unentangled states satisfy $\det(\Psi) = 0$, which is a zero-measure condition, thus a randomly chosen state in $\mathcal{H} \otimes \mathcal{H}$ is entangled with probability 1.

Question 2: Quantum Circuits

(a)

(i) Fredkin gate's action is like:

$$\begin{aligned}
|000\rangle &\to |000\rangle \\
|001\rangle &\to |001\rangle \\
|010\rangle &\to |010\rangle \\
|011\rangle &\to |101\rangle \\
|100\rangle &\to |100\rangle \\
|101\rangle &\to |011\rangle \\
|110\rangle &\to |110\rangle \\
|111\rangle &\to |111\rangle
\end{aligned}$$

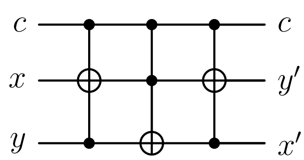
(ii) The matrix has 1s on its main diagonal, except that the positions of rows 3 and 5 are swapped, and the rest remain unchanged.

$$M_F = \begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}$$

(iii) Let the inputs be x and y, corresponding to the 1st and 3rd qubits of the Fredkin gate, respectively, and set the 2nd input qubit to $|0\rangle$. Then the 2nd output qubit gives the result of the AND operation.

$$\begin{aligned}
|000\rangle \to |000\rangle &\iff |00\rangle \to |0\rangle \\
|001\rangle \to |001\rangle &\iff |01\rangle \to |0\rangle \\
|100\rangle \to |100\rangle &\iff |10\rangle \to |0\rangle \\
|101\rangle \to |011\rangle &\iff |11\rangle \to |1\rangle
\end{aligned}$$

(iv) The Fredkin gate (CSWAP) is constructed by decomposing the SWAP operation into 3 CNOTs, and then adding a control to each to form 3 Toffoli gates.



- **Gate 1:** Toffoli(c, x, y) performs $y \leftarrow y \oplus cx$.
- **Gate 2:** Toffoli(c, y, x) performs $x \leftarrow x \oplus cy$.
- **Gate 3:** Toffoli(c, x, y) performs $y \leftarrow y \oplus cx$.

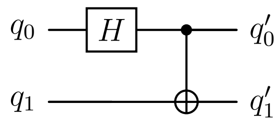
This sequence swaps the target bits x and y if and only if the control bit $c = 1$.

(b)

A 2-qubit circuit that performs

$$
\begin{aligned}
|00\rangle &\longmapsto \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big) = |\Phi^+\rangle\,,\\
|01\rangle &\longmapsto \frac{1}{\sqrt{2}}\big(|01\rangle + |10\rangle\big) = |\Psi^+\rangle\,,\\
|10\rangle &\longmapsto \frac{1}{\sqrt{2}}\big(|00\rangle - |11\rangle\big) = |\Phi^-\rangle\,,\\
|11\rangle &\longmapsto \frac{1}{\sqrt{2}}\big(|01\rangle - |10\rangle\big) = |\Psi^-\rangle\,.
\end{aligned}
$$

These four outputs are exactly **the four Bell states**, and the transformation is the well-known "Bell–basis encoder". It can be realised with only two gates:



In the computational basis $\big\{|00\rangle\,, |01\rangle\,, |10\rangle\,, |11\rangle\big\}$ the circuit unitary is

$$
U = \mathrm{CNOT}\,(H \otimes I) = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix}.
$$

Applying U to each basis vector reproduces the required mappings, e.g.

$$
U\,|00\rangle = \frac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big), \quad U\,|11\rangle = \frac{1}{\sqrt{2}}\big(|01\rangle - |10\rangle\big),
$$

and similarly for $|01\rangle$, $|10\rangle$. Hence the circuit above solves the problem.

Question 3: Analysis of Grover's Algorithm

(a)

Let

$$
|\psi\rangle = a\,|\alpha\rangle + b\,|\beta\rangle
$$

be an arbitrary state in the subspace spanned by $|\alpha\rangle$ and $|\beta\rangle$. The quantum oracle O_f acts as

$$
O_f\,|x\rangle = \begin{cases} -\,|x\rangle\,, & \text{if } f(x) = 1 \text{ (i.e., } x \text{ is a solution)},\\ |x\rangle\,, & \text{if } f(x) = 0. \end{cases}
$$

Thus,

$$
O_f\,|\alpha\rangle = |\alpha\rangle\,, \quad O_f\,|\beta\rangle = -\,|\beta\rangle\,.
$$

Hence,

$$O_f |\psi\rangle = a |\alpha\rangle - b |\beta\rangle \in \text{span}\{|\alpha\rangle, |\beta\rangle\}.$$

Next, the Grover diffusion operator is $G = 2|s\rangle\langle s| - I$, where $|s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$. Note that $|s\rangle$ can be expressed as

$$|s\rangle = \sqrt{\frac{N-M}{N}} |\alpha\rangle + \sqrt{\frac{M}{N}} |\beta\rangle.$$

Therefore,

$$G = 2|s\rangle\langle s| - I$$

is a linear combination of projectors involving only $|\alpha\rangle$ and $|\beta\rangle$. Since $|\psi\rangle \in \text{span}\{|\alpha\rangle, |\beta\rangle\}$, applying G to $|\psi\rangle$ yields another vector in the same subspace. Thus, both O_f and G preserve the two-dimensional subspace.

(b)

Because the entire evolution is confined to the $\{|\alpha\rangle, |\beta\rangle\}$ subspace, we analyze the algorithm geometrically in this plane. Define the angle θ via

$$\sin\theta = \langle\beta\rangle\, s = \sqrt{\frac{M}{N}}, \quad \cos\theta = \langle\alpha\rangle\, s = \sqrt{\frac{N-M}{N}}.$$

The initial state is $|s\rangle = \cos\theta\,|\alpha\rangle + \sin\theta\,|\beta\rangle$, and each Grover iteration GO_f rotates the state vector by 2θ toward $|\beta\rangle$.

To reach a state close to $|\beta\rangle$, we need approximately R iterations such that

$$(2R+1)\theta \approx \frac{\pi}{2}.$$

For small θ (i.e., $M \ll N$), $\theta \approx \sqrt{M/N}$, so

$$R \approx \frac{\pi}{4\theta} \approx \frac{\pi}{4}\sqrt{\frac{N}{M}}.$$

Therefore, the query complexity (number of oracle calls) is

$$\mathcal{O}\left(\sqrt{\frac{N}{M}}\right),$$

which generalizes the standard $\mathcal{O}(\sqrt{N})$ result for $M = 1$.